ASSIGNMENT 2 · DATA PREPARATION & BASELINE · BRIEF RESEARCH REPORT

Which public-procurement contracts fail? Evidence from 359 thousand procurements in Kazakhstan

Aziz Abdulin · October 2026 · Code & notebook: github.com/awzwz/tender-risk-ml (notebooks/02_data_preparation.ipynb)

1. Problem

Kazakhstan's e-procurement portal handles about 1.8 million procurements a year. Some of the resulting contracts are terminated or never signed, which forces repeated procedures, delays delivery and wastes public money. Oversight bodies can audit only a small fraction of procurements, so they need to know *in advance* where failures are likely. Existing risk-scoring approaches rely mostly on expert "red flags" such as single bidding (Fazekas et al., 2016; Fazekas & Kocsis, 2020). This study asks which characteristics known at the moment of award are associated with contract failure (RQ1: method and competition; RQ2: value, subject, region) and whether they predict failure on a later period (RQ3).

2. Data

The data come from the official open-data API of the portal (OWS v3, Ministry of Finance of the Republic of Kazakhstan). All 630,762 procurements published between 1 January and 31 March 2024 were collected together with their 1.29 million lots, 1.07 million supplier applications, 556 thousand contract records and the register of unreliable suppliers (RNU). Q1 2024 was chosen so that every contract has had more than two years to reach a final status. The raw snapshot is published as a GitHub release, which makes the notebook fully reproducible without an API token.

3. Methodology

Data quality was assessed per table (missing values, duplicates, wrong types, invalid values, outliers, inconsistent categories, referential integrity). The key cleaning step was collapsing contract amendments: 13.6 % of contract rows are later versions of the same contract, and only the last version carries the final status. The unit of analysis is a procurement that produced at least one contract (n = 358,789). The target **contract_failed** equals 1 if any of its contracts ended as *terminated* (unilaterally or by agreement) or *not concluded*. Twenty features describing the procedure, its value, lots, competition and organiser were built using information available at award time only; a leakage audit excluded execution fields, system timestamps, procurement final status and the RNU register, whose entries are a consequence of termination (Kaufman et al., 2012). Data were split chronologically (train: Jan–Feb, validation: 1–15 Mar, test: 16–31 Mar; 74/15/11 %). A class-weighted logistic regression served as the baseline; because classes are imbalanced, it was evaluated with ROC-AUC, PR-AUC and F1 (Saito & Rehmsmeier, 2015).

4. Results

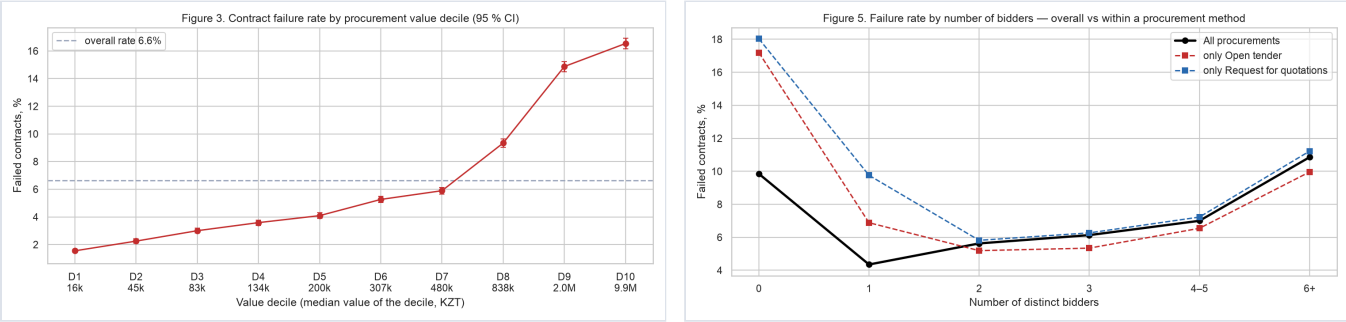


Figure 1 (Figures 3 and 5 of the notebook). Contract failure rate by procurement-value decile (left) and by number of bidders, overall and within a procurement method (right). Source: author's calculations, OWS v3, Q1 2024.

- **Base rate.** 6.6 % of procurements end with a failed contract (4.5 % under a strict definition without terminations by agreement); the most common failure is unilateral termination. The rate is stable across weeks (6.4–6.8 % per month).
- **RQ1 — method and competition.** Automatic lowest-price requests for quotations fail most often (9.0 %), single-source purchases after a failed tender 3.2 %, framework call-offs 1.2 %. Single-bidder procurements look safest overall (4.4 %), but this is a Simpson's paradox: within the same method one bidder is riskier than two (e.g. 9.8 % vs 5.8 % for requests for quotations), supporting H1. The effect of competition is U-

shaped — with 6+ bidders failure rises to 11 %, consistent with the winner's curse in price competition (Thaler, 1988).

- **RQ2 — value, subject, region.** Failure rises tenfold from the cheapest to the most expensive value decile (1.5 % → 16.5 %), supporting H2. Goods fail twice as often as services (9.1 % vs 4.2 %); Astana, Shymkent and Almaty have roughly twice the rate of several regions (10.6 % vs ≈ 5 %).
- **RQ3 — predictability.** The baseline generalises to the unseen test period (table), supporting H3: half of the 1 % of procurements it ranks as most risky actually failed, 7.6 times the base rate.

Model (test period, 16–31 March 2024)	ROC-AUC	PR-AUC	F1	Precision in top 1 %
Prior classifier (no model)	0.500	0.066	—	0.057
Logistic regression (class-weighted, threshold tuned on validation)	0.792	0.262	0.324	0.504

5. Limitations

The sample covers a single quarter, so seasonal effects are not represented. The label combines supplier failures with customer decisions (termination by agreement); a strict variant is stored for sensitivity analysis. Track records of suppliers and organisers cannot be used without leakage because the termination dates of earlier contracts are not observed. About 16 % of procurements have no priced offers and some offers use inconsistent units. The findings are associational, not causal.

6. Conclusions

Contract failure in Kazakhstan's procurement is neither rare nor random: it concentrates in large, goods-type, lowest-price procedures and in the largest cities, and it depends on competition in a U-shaped way. The popular "single bidder" red flag is meaningful only relative to the procurement method. Information available at award time already ranks risky procurements far better than chance, which justifies the next stage of the project: tree-based models able to capture the interactions found here, and text features from lot descriptions.

References

Fazekas, M., & Kocsis, G. (2020). Uncovering high-level corruption: Cross-national objective corruption risk indicators using public procurement data. *British Journal of Political Science, 50*(1), 155–164.

Fazekas, M., Tóth, I. J., & King, L. P. (2016). An objective corruption risk index using public procurement data. *European Journal on Criminal Policy and Research, 22*(3), 369–397.

Kaufman, S., Rosset, S., Perlich, C., & Stitelman, O. (2012). Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data, 6*(4), 1–21.

Ministry of Finance of the Republic of Kazakhstan. (2026). *Public procurement portal — open data API (OWS v3)*. https://ows.goszakup.gov.kz (accessed 6 October 2026).

Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLoS ONE, 10*(3), e0118432.

Thaler, R. H. (1988). Anomalies: The winner's curse. *Journal of Economic Perspectives, 2*(1), 191–202.